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Cover device for photoresist container

Abstract

A cover device includes a guide module which is coupled to an upper end portion of the container, has an upward inclination at a predetermined angle, has an inclined protruding unit that protrudes, and is formed with a communication port through which the inside and the outside of the housing space communicate; a discharge module in which one side along a longitudinal direction penetrates the communication port and is disposed inside the container, the other side is exposed to the outside to form a discharge port, and an intermediate portion along the longitudinal direction has an expansion fixing unit which is expanded, at least a part of periphery is in contact with an inner surface of the communication port, and is inserted and fixed; and a fixing module which is fixedly coupled to the inclined protruding unit in a state in which the other part of the discharge module penetrates.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5083683	12/1991	Knickerbocker	222/464.7	B05B 11/0037
5586589	12/1995	Voelker	141/DIG.1	B67D 7/344

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2016-159981	12/2015	JP	N/A
10-2004-0017479	12/2003	KR	N/A
10-1350282	12/2013	KR	N/A
10-2023-0028085	12/2022	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims under 35 U.S.C. § 119(a) the benefit of Korean Patent Application No. 10-2023-0056524 filed on Apr. 28, 2023, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(a) Technical Field

(2) The present invention relates to a cover device for a photoresist container, which is coupled to a container into which a gas is injected to discharge a photoresist aqueous solution inside, and prevents unintentional separation due to a pressure.

(b) Background Art

(3) In general, a photolithography process includes a process of positioning a wafer, which has undergone a series of semiconductor manufacturing processes, on a spin chuck of a coating device and coating it with a photoresist.

(4) Here, the photoresist includes a resin, an inhibitor as a sensitizer activated by light, and a

solvent of a liquid state. The photoresist supplied to the photoresist supply line is supplied to a photoresist supply line through a photoresist supply pipe (Teflon tube) coupled to a container cap while being housed in the container, and the photoresist thus supplied to photoresist supply line is supplied to a coating device via a plurality of valves and an intermediate tank.

(5) Here, a container for supplying the photoresist aqueous solution and a cover device combined thereto are used, and configured to supply a high-pressure gas into the container and discharge the photoresist aqueous solution therein and transfer it to the supply pipe.

(6) At this time, the container is provided with a cover device for discharging the photoresist aqueous solution and for supplying high-pressure gas to the inside of the container. However, because the inside of the container is maintained in a high-pressure state, a problem arises that a part of the cover device is separated.

(7) In particular, if a part of the cover device is unintentionally separated, the user may be shot by the separated part, or the photoresist aqueous solution inside may be partially scattered on the user's body, causing a safety accident.

(8) Therefore, a development of technique for solving such these problems is necessary.

PRIOR ART DOCUMENTS

Patent Document

(9) (Patent Document 1) Korean Laid-open Patent Publication No. 10-2004-0017479

SUMMARY OF THE INVENTION

(10) The present invention was devised to solve the above-described problems of the related art, and object thereof is to provide a cover device for a photoresist container that is coupled to a container for injecting a gas at high pressure and discharging a photoresist aqueous solution therein, and enables the gas to be discharged, when the cover device is partially separated due to pressure.

(11) The objects of the present invention are not limited to the above objects, and other objects not mentioned will be clearly understood by those skilled in the art from the following description.

(12) The present invention to achieve the above object relates to a cover device which is coupled to an opening portion of a container in which a photoresist aqueous solution is housed to seal the inside and simultaneously discharge the photoresist aqueous solution to the outside, the cover device including: a guide module which is coupled to an upper end portion of the container, has an upward inclination at a predetermined angle, has an inclined protruding unit that protrudes, and is formed with a communication port through which the inside and the outside of the housing space communicate; a discharge module which is formed in a long pipe shape, and in which one side along a longitudinal direction penetrates the communication port and is disposed inside the container, the other side is exposed to the outside to form a discharge port, and an intermediate portion along the longitudinal direction has an expansion fixing unit which is formed to expand, is in contact with an inner surface of the communication port on at least a part of periphery, and is inserted and fixed; and a fixing module which is fixedly coupled to the inclined protruding unit in a state in which the other part of the discharge module penetrates, and is formed so that the discharge module is pressed toward the guide module, wherein the communication port is formed to have an inner diameter that decreases toward the inside of the container in at least a partial region and is fixed by pressing.

(13) In addition, the discharge module may further include a channel unit which is formed long, has a part disposed inside the container, and has a discharge channel formed therein, and a connecting unit which protrudes to extend from the expansion fixing unit, is exposed to outside, and has a discharge port extending with the discharge channel.

(14) Also, the inclined protruding unit may further include a protrusion pressing part formed in at least a part along the inner surface of the communication port, protrudes toward the center, and comes into contact with the expansion fixing unit.

(15) Also, the protrusion pressing part may be formed with a first inclined surface that protrudes at

a predetermined angle along the longitudinal direction.

(16) Further, the inclined protruding unit may be formed with a second inclined surface, which is continuous with the first inclined surface and is inclined in an opposite direction inside the communication hole, and may have a relatively larger inclination angle than the first inclined surface.

(17) In addition, the inclined protruding unit may further include a decompression part which is formed in at least a part along the periphery, and when the expansion fixing unit is pushed out from the inside of the communication port by a preset length or more and slides toward the connecting unit, discharges the gas inside the container to the outside.

(18) Also, the decompression part may be formed in a hole shape through which inside and outside communicate in at least a part along the periphery of the communication port, and may be formed to be opened when the expansion fixing unit moves rearward.

(19) Also, the decompression part may be formed in a slit shape at an upper end portion of the inclined protruding unit, and opened when the expansion fixing unit moves rearward.

(20) In addition, the fixing module may include a coupling unit through which the connecting unit penetrates, which wraps around and fixes the inclined protrusion in a state of being in contact with the expansion fixing unit, and has an annular latch formed on the upper end of an outer surface; and an adjusting unit which is rotatably coupled to an outer peripheral surface of the coupling unit, has a plurality of elastic pieces formed to correspond to the latch to rotate the coupling unit within a preset level, and adjusts the expansion fixing unit to be pressed toward the container.

(21) In addition, the fixing module may include a coupling unit, through which the other part of the discharge module penetrates, and which surrounds at least a part of the inclined protruding unit in a state of being in contact with the expansion fixing unit; and an adjusting unit which has a fixed length, and an intermediate portion fixed to the coupling unit, and in which one side along the longitudinal direction selectively comes into contact with the inclined protruding unit to adjust coupling with the coupling unit, wherein the adjusting unit is operated to press the expansion fixing unit in the direction of the container, and the expansion fixing unit may be prevented from being detached in the opposite direction.

(22) In addition, the inclined protrusion unit may include a serrated locking protrusion on at least a part of the outer surface, and a plurality of protrusions are continuously formed along the protruding longitudinal direction, and the adjusting unit may be coupled to at least a part of the locking protrusion and be selectively fixed.

(23) The cover device for photoresist container of the present invention for solving the above problems has the following effects.

(24) The cover device according to the present invention includes a guide module having a protruding communication port, a discharge module which penetrates the communication port to discharge the photoresist aqueous solution to the outside, and a fixing module which maintains a coupling between the discharge module and the guide module. The discharge module is inserted and fixed into the container, and an inner diameter of the communication port is configured to decrease along the longitudinal direction, so that the expansion fixing unit can be stably fixed and sealed as it is pressed toward the communication port.

(25) The effects of the present invention are not limited to the effects described above, and other effects not mentioned will be clearly understood by those skilled in the art from the description of the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a cover device for a photoresist container according to the

present invention;

(2) FIG. 2 is an exploded perspective view showing a detailed configuration of the cover device for the photoresist container according to the present invention;

(3) FIG. 3 is a cross-sectional view showing an internal structure of the cover device for the photoresist container of the present invention;

(4) FIGS. 4A and 4B are plan views showing a configuration of a fixing module in the cover device for the photoresist container of FIG. 1;

(5) FIG. 5 is a diagram showing a detailed configuration of a communication port in the cover device for the photoresist container of FIG. 1;

(6) FIGS. 6A and 6B are diagrams showing a configuration of a depressurization unit in the cover device for the photoresist container of FIG. 1; and

(7) FIG. 7 is a diagram showing a modified form of the depressurization unit in the cover device for the photoresist container of FIG. 1.

DETAILED DESCRIPTION

(8) Preferred embodiments of the present invention, which can specifically achieve the object of the present invention, will be described below with reference to the accompanying drawings. In describing this embodiment, the same names and the same reference numerals are used for the same configurations, and the additional description associated therewith will be omitted.

(9) In addition, in describing the embodiments of the present invention, it will be clarified in advance that the components having the same functions use only the same names and the same reference numerals, but are not substantially the same as the conventional ones.

(10) Moreover, the terms used in the embodiments of the present invention are merely used to describe particular embodiments and are not intended to be limiting of the present invention.

Singular expressions include plural expressions unless they are clearly different contextually.

(11) Further, it is to be understood that, in the embodiments of the present invention, terms such as “including” or “having” are intended to specify the presence of the features, numbers, steps, operations, constituent elements, components or combinations thereof described herein, but do not preclude the presence or the possibility of addition of other features, numbers, steps, operations, constituent elements, components, or combinations thereof.

(12) FIG. 1 is a diagram showing an external form of a cosmetic case according to the present invention.

(13) First, a cover device for a photoresist container **10** according to the present invention will be schematically described below with reference to FIGS. 1 to 4.

(14) FIG. 1 is a perspective view of the cover device for the photoresist container **10** of the present invention, and FIG. 2 is an exploded perspective view showing the detailed configuration of the cover device for the photoresist container **10** of the present invention.

(15) FIG. 3 is a cross-sectional view showing an internal structure of the cover device for the photoresist container **10** of the present invention, and FIGS. 4A and 4B are plan views showing a configuration of a fixing module **300** in the cover device for the photoresist container **10** of FIG. 1.

(16) The cover device of the photoresist container **10** according to the present invention is sealed by being coupled to an upper part of the container **10** in which the photoresist aqueous solution is contained, and simultaneously a separate high-pressure gas is introduced from the outside, and the photoresist aqueous solution is supplied to the outside by the pressure accordingly.

(17) Referring to the shown drawings, the cover device according to the present invention roughly includes a guide module **100**, a discharge module **200** and a fixing module **300**, and the guide module **100** is coupled to an opened portion of the container **10** to seal the inside of the housing space **12**, and simultaneously enables the photoresist aqueous solution to be discharged to the outside through a separate communication port **112**.

(18) Here, at least a part of the container **10** is formed to extend upward, is formed to protrude while having an upward inclination at a preset angle, and is coupled to the fixing module **300**,

which will be described below.

(19) Specifically, the guide module **100** includes an inclined protruding unit **110** that is disposed at the open upper end of the container **10** and protrudes with an upward inclination, and a cover unit **120** fixedly coupled to the container **10** in the state of being connected to the cover unit **120**.

(20) The inclined protruding unit **110** is made of synthetic resin, is formed in a pipe shape, has a form protruding by a certain length, and has a communication port **112** communicating with the container **10** formed therein. At this time, the inclined protruding unit **110** includes serrated locking protrusions **113** on at least a part of the outer surface thereof, a plurality of protrusions **113** are continuously formed along a protruding longitudinal direction, and the fixing module **300**, which will be described below is combined thereto.

(21) The inclined protruding unit **110** has a lower end portion is formed to be able to make sealing contact corresponding to the shape of the opening of the container **10**, and at least a portion of the inclined protruding unit **110** is coupled to the cover unit **120**.

(22) In this embodiment, the inclined protruding unit **110** has a T-shape as shown, has an extended part **114** having an extended periphery at its lower end, and has a form protruding long in the upward direction. At this time, the communication port **112** is elongated in a vertical direction and communicates with the inside of the container **10**.

(23) In addition, the cover unit **120** is configured to maintain a combined state without detaching in the state in which the inclined protruding unit **110** and the container **10** are in contact with each other, and is fixedly coupled to the opening in the state of being connected to the extended part **114**.

(24) Specifically, the cover unit **120** is formed in a ring shape, through which the inclined protruding unit **110** penetrates, at least a part of which wraps around the extended part **114**, and is coupled to the container **10**. At this time, separate screw threads are formed on the container **10**, and the lower side of the cover unit **120** wraps around the container **10** and is screwed and fixed to the screw threads.

(25) Here, the cover unit **120** has a ring-shaped central hole through which a part of the inclined protruding unit **110** penetrates, but the hole is relatively smaller than the extended part **114**. Accordingly, when the cover unit **120** is coupled to the container **10**, the inclined protruding unit **110** can maintain the combined state without being detached.

(26) In this embodiment, the cover unit **120** is formed of two members that are stacked in the vertical direction, the extended part **114** is disposed between the members, and the cover unit **120** and the inclined protruding unit **110** can maintain the combined state.

(27) In this way, the guide module **100** includes the inclined protruding unit **110** and the cover unit **120**, is fixedly coupled to the opening of the container **10**, and allows the inclined protruding unit **110** to be disposed while protruding upward at a preset angle.

(28) Additionally, the inclined protruding unit **110** according to the present invention has a separate protrusion pressing part **130** formed along the inner surface of the communication hole **112**, and a part of the discharge module **200** is inserted into the communication hole **112** and pressed by the protrusion pressing part **130**.

(29) A detailed configuration of the protrusion pressing part **130** will be described below with reference to FIG. 5.

(30) On the other hand, the discharge module **200** is formed in a long pipe shape, passes through the communication port **112**, has one side inserted into the housing space **12** of the container **10**, and the other side exposed to the outside to form a discharge port **232**.

(31) Specifically, the discharge module **200** is combined and fixed in a state of passing through the guide module **100**, one side is disposed in the housing space **12**, and the other side is disposed outside the container **10** to discharge the contained photoresist aqueous solution to the outside.

(32) At this time, when the discharge module **200** is coupled to the guide module **100**, the discharge module **200** is tightly fixed and inserted to form a discharge channel and an air channel, and the discharge channel is configured to abut on the bottom surface of the housing space **12** and

discharge the photoresist aqueous solution to the outside. The rest of the air channel communicates with the upper portion of the housing space **12** and is supplied with a high-pressure gas so that the photoresist aqueous solution can be discharged to the outside by pressure.

(33) In the present invention, a part of the discharge module **200** has a periphery corresponding to the periphery of the communication port **112** along the longitudinal direction, and the discharge module **200** is inserted into the inner surface of the communication port **112**, and roughly includes a channel unit **210**, an expansion fixing unit **220** and a connecting unit **230**.

(34) The channel unit **210** is formed long, and a part thereof is disposed inside the container **10**, and a discharge channel is formed therein to discharge the photoresist aqueous solution to the other side.

(35) Specifically, the channel unit **210** is formed long in a general pipe form, and is disposed so that one side is in contact with the bottom surface of the container **10**. At this time, as shown, the channel unit **210** is disposed in a state of being inclined at a certain angle by the guide module **100**, and is seated so that one side is in contact with the edge portion of the bottom surface in the housing space **12** to completely discharge the remaining photoresist aqueous solution.

(36) In this embodiment, the channel unit **210** is formed long and relatively narrower than the opening of the container **10**, and a discharge channel is formed inside.

(37) On the other hand, the expansion fixing unit **220** is formed to extend to the other side of the channel unit **210** and is configured to be inserted and fixed into the communication port **112** so that a partial region along the longitudinal direction of the channel unit **210** is expanded, and the discharge channel is formed to extend inside.

(38) Specifically, the expansion fixing unit **220** preferably has a periphery and a length corresponding to the periphery of the communication port **112**, and is inserted through the communication port **112** when inserting the channel unit **210**, and is configured to seal the communication port **112**.

(39) At this time, the expansion fixing unit **220** has an air channel separate from the discharge channel, and is connected to the connecting unit **230** to be described later to supply high-pressure air to the inside of the housing space **12**.

(40) In this embodiment, the expansion fixing unit **220** is formed in a cylindrical shape corresponding to the shape of the communication port **112**, and is provided with at least one or more sealing along the periphery to prevent unintentional flowing of fluid when coming into contact with the communication port **112**.

(41) Here, the expansion fixing unit **220** is configured so that the protrusion pressing part **130** to be described below comes into contact with the side surface along the periphery and is pressurized in a tightening manner when inserted into the communication port **112** to maintain the combined state stably.

(42) On the other hand, the connecting unit **230** is provided on the other side of the channel unit **210** with the expansion fixing unit **220** at the center, and is formed so that the air channel and the discharge channel are independently connected.

(43) More specifically, the connecting unit **230** extends and protrudes from the expansion fixing unit **220** to be exposed to the outside as shown, and includes a discharge port **232** connected to the discharge channel, and a supply part **234** connected to the air channel. At this time, the discharge port **232** is connected to a device for supplying the photoresist aqueous solution, the supply part **234** is connected to means for supplying high-pressure air supplied into the housing space **12**, and a nitrogen gas is supplied as the high-pressure air in this embodiment.

(44) In this embodiment, the connecting unit **230** is configured to pass through the fixing module **300** to be described below and exposed to the outside as shown, and the discharge port **232** and the supply portion **234** are provided independently.

(45) In this way, the discharge module **200** includes the channel unit **210**, the expansion fixing unit **220** and the connecting unit **230**, and the expansion fixing unit **220** is coupled to communication

port **112** with one side penetrating the guide module **100**.

(46) On the other hand, the fixing module **300** is fixedly coupled to the inclined protruding unit **110** with the other part of the discharge module **200** penetrating therethrough so that the discharge module **200** is pressed toward the guide module **100**.

(47) Specifically, the fixing module **300** is configured to wrap a part of the other side of the discharge module **200** and simultaneously be fixedly coupled to the inclined protruding unit **110** to press the expansion fixing unit **220**.

(48) At this time, the fixing module **300** is formed in the shape of a ring in which a hole relatively smaller in size than the periphery of the expansion fixing unit **220** is formed at the center of the upper end of the fixing module **300**, the connecting unit **230** is configured to pass therethrough, and the fixing module **300** is formed to wrap the outer surface of the inclined protruding unit **110**.

(49) In addition, the inclined protruding unit **110** includes a serrated locking protrusion **113** on at least a part of the outer surface thereof, a plurality of protrusions **113** are continuously formed along the protruding longitudinal direction, the fixing module **330** wraps around the inclined protruding unit **110**, and at least a part thereof is configured to be in selectively contact with and be fixed to the locking protrusion **113**.

(50) As a result, the fixing module **300** moves along the longitudinal direction of the inclined protruding unit **110** toward the container **10**, and is fixed to the locking protrusion **113** during movement, thereby preventing the fixing module **300** from unintentionally detaching, and the expansion fixing unit **220** can be pressed into the communication port **112** to maintain the combined state.

(51) Here, the inclined protruding unit **110** according to the present invention is formed such that the inner diameter decreases along the direction of the container **10** in at least a partial region of the communication port **112**. When the fixing module **300** presses the expansion fixing unit **220** into the communication port **112**, the expansion fixing unit **220** is pressed along the direction in which the inner diameter of the communication port **112** decreases, and the communication port **112** can be stably sealed.

(52) In this embodiment, the fixing module **300** roughly includes a coupling unit **310** and an adjusting unit **320**, and the coupling unit **310** is configured to wrap around the outer surface of the inclined protruding unit **110** and press the expansion fixing unit **220**.

(53) Specifically, the coupling unit **310** is penetrated by the connecting unit **230**, wraps around and fixes the inclined protruding unit **110** while being in contact with the expansion fixing unit **220**, and is fixed to the inclined protruding unit **110** by the adjusting unit **320**.

(54) Here, as shown in FIGS. 4A and 4B, the coupling unit **310** has a cylindrical shape with a constant thickness, an upper part thereof is formed to have a relatively small inner diameter, and wraps the expansion fixing unit **220** to prevent from escaping upward in the state in which the channel unit **210** penetrates.

(55) And, it is coupled to wrap the upper end portion of the inclined protruding unit **110**, and is configured so that the position can be adjusted toward the container **10**.

(56) That is, the coupling unit **310** is coupled to the inclination protruding unit **110** in the form that wraps the expansion fixing unit **220** from above while the discharge module **200** is inserted through the inclination protruding unit **110**, to prevent detachment of the discharge unit **200** and simultaneously adjust the depth of insertion.

(57) At this time, the coupling unit **310** is configured to be coupled to the adjusting unit **320** to be described later, and selectively fixed or separated from the inclined protruding unit **110** through adjustment of the adjusting unit **320**.

(58) In this embodiment, the coupling unit **310** has a cylindrical shape as shown, and a coupling groove **312** is formed through a part of the side surface along the periphery, a part of the outer surface of the inclined protruding unit **110** is exposed, the adjusting unit **320** is coupled to the coupling groove **312** to be in selectively contact with and fixed to the outer surface of the inclined

protruding unit **110**.

(59) On the other hand, the adjusting unit **320** is provided on the coupling unit **310** to selectively adjust the movement of the coupling unit **310**, has a certain length, and is fixed to the coupling unit **310** at its intermediate portion. One side along the longitudinal direction comes into selectively contact with the inclined protruding unit **110** to adjust whether it is coupled to the coupling unit **310**.

(60) At this time, the adjusting unit **320** prevents the expansion fixing unit **220** from being unintentionally separated in the opposite direction while the expansion fixing unit **220** is pushed toward the container **10**, and selectively separate the coupling unit **310** by the user's operation.

(61) Specifically, at least one or more adjusting units **320** are provided along the periphery of the coupling unit **310**, and is disposed vertically while being seated in the coupling groove **312**, and the intermediate portion along the longitudinal direction is rotatably coupled to the coupling unit **310**.

One end portion along the longitudinal direction is formed to correspond to the shape of the locking protrusion **113** and is selectively brought into contact therewith. Here, one end portion along the longitudinal direction of the adjusting unit **320** has at least one or more teeth, and is coupled to at least a part in the region of the locking protrusion **113** exposed through the coupling groove **312** to fix the coupling unit **310**.

(62) As a result, when one side of the adjusting unit **320** along the longitudinal direction comes into contact with the locking protrusion **113**, the coupling unit **310** moves toward the container **10** or can remain a coupled state, without detaching from the inclined protruding unit **110**.

(63) In contrast, as shown in FIGS. **4A** and **4B**, when the other side of the adjusting unit **320** is pushed to separate one side from the locking protrusion **113**, the coupling unit **310** can be in a state of being separable from the inclined protruding unit **110**.

(64) In this embodiment, a plurality of adjusting units **320** are disposed along the periphery of the coupling unit **310** at regular intervals, and when the user presses and operates the other side of the adjusting unit **320** at the same time, one side can be maintained selectively in a state of being separated from or in contact with the locking protrusion **113**.

(65) At this time, the adjusting unit **320** is configured such that the intermediate portion along the longitudinal direction rotates around the coupling unit **310** with a rotating axis, and is configured such that a restoring force pressed toward the inclined protruding unit **110** acts on one side. That is, the adjusting unit **320** is rotatably coupled to the coupling unit **310** and is provided with a restoring force through a separate elastic member (not shown), so that one side may maintain a contact state with the locking protrusion **113** when the user does not press the other side.

(66) Thus, the fixing module **300** according to the present invention includes the coupling unit **310** and the adjusting unit **320**. When the user presses the coupling unit **310** toward the containing container **10** in the state of being coupled to the inclined protruding unit **110** to move the expansion fixing unit **220** and the channel unit **210** together toward the containing container **10**, the lower end portion of the channel unit **210** can be disposed close to the bottom surface of the containing container **10**.

(67) That is, the fixing module **300** is pressed and fixed by the coupling unit **310** in a state in which the discharge module **200** penetrates the communication port **112**, and when the coupling unit **310** presses the discharge module **200** toward the container **10**, the locking protrusion **113** and one side of the adjusting unit **320** are brought into contact with each other to adjust the insertion depth step by step. Accordingly, the discharge module **200** can be prevented from being inserted into the container **10** above a certain level.

(68) As described above, the cover device according to the present invention includes the guide module **100**, the discharge module **200** and the fixing module **300**, the discharge module **200** is inserted and fixed into the container **10**, and the inner diameter of the communication port **112** is configured to decrease along the longitudinal direction. Accordingly, the fixing and sealing can be stably performed as the expansion fixing unit **220** is pressed toward the communication port **112**.

(69) Next, with reference to FIGS. 5 to 7, the detailed configuration of the inclined protruding unit **110** according to the present invention will be described as follows.

(70) FIG. 5 is a diagram showing the detailed configuration of the communication port **112** in the cover device of the photoresist container **10** of FIG. 1, FIGS. 6A and 6B are diagrams showing the configuration of the decompression unit **140** in the cover device of the photoresist container **10** of FIG. 1, and FIG. 7 is a diagram showing a modified form of the decompression unit **140** in the cover device of the photoresist container **10** of FIG. 1.

(71) First, referring to the drawings, the inclined protruding unit **110** has a pipe-shaped communication hole **112** formed at the center, and further includes a protrusion pressing part **130** that protrudes toward the center along the inner surface of the communication hole **112** and makes contact with the expansion fixing unit **220**, and a decompression part **140** that discharges the air inside the housing space **12** to the outside at the time of an emergency.

(72) The protrusion pressing part **130** is formed in a general protrusion shape or ring shape, and is configured such that the inner diameter decreases along the longitudinal direction of the communication port **112**. Here, when the expansion fixing unit **220** is inserted, the protrusion pressing part **130** is configured to press the side surface to suppress the detachment, and the pressing strength of the side surface also increases, as the expansion fixing unit **220** is pressed and inserted.

(73) As a result, as the fixing module **300** presses the expansion fixing unit **220** toward the communication port **112**, the protrusion pressing part **130** can also press the side surface of the expansion fixing unit **220** to suppress the detachment as much as possible.

(74) In this embodiment, the protrusion pressing part **130** is formed in a ring shape as shown and is formed on the inner surface of at least a part of the communication port **112**, and includes a first inclined surface **132** that is formed to protrude along the longitudinal direction at a certain angle, and a second inclined surface **134** which is contiguous with the first inclined surface **132** and has an inclination in the opposite direction.

(75) The first inclined surface **132** is formed on the inner surface of the communication port **112** at a preset angle so that the inner surface gradually decreases along the longitudinal direction, and the second inclined surface **134** is configured to have an inclination in the opposite direction to the first inclined surface **132**. At this time, the second inclined surface **134** is configured to have a relatively larger inclination angle than the first inclined surface **132**, and when the expansion fixing unit **220** is inserted into the communication hole **112**, the corner portion is configured to press the side surface of the expansion fixing unit **220**.

(76) As a result, when the expansion fixing unit **220** is inserted into the communication port **112**, it is pressed by the first inclined surface **132**, the pressing of the side surfaces increases, and can be suppressed from sliding in the opposite direction by the second inclined surface **134**.

(77) In this way, the inclined protruding unit **110** according to the present invention includes the protrusion pressing part **130** on the inner surface of the communication hole **112**, thereby enabling complete sealing when inserting the expansion fixing unit **220**, and suppressing detachment in the opposite direction at the same time.

(78) Next, the inclined protruding unit **110** includes a separate decompression part **140**, as shown in FIGS. 6A and 6B.

(79) The decompression part **140** is formed along at least a part of the periphery of the inclined protruding unit **110**, and when the expansion fixing unit **220** is pushed out more than a preset length inside the communication port **112** and slides in the direction of the connecting unit **230**, the gas inside the container **10** is discharged to the outside.

(80) Specifically, at least one or more decompression parts **140** are provided, are formed in the shape of a hole or a slit on the inner surface of the communication port **112** to cause the inside and the outside to communicate with each other, and are selectively exposed to the outside according to the inserted position of the expansion fixing unit **220**.

(81) At this time, as shown in FIGS. 6A and 6B, the decompression part **140** is formed in a hole-like shape in which the inside and the outside communicate at least partially along the periphery of the communication hole **112**, and is formed to be open when the expansion fixing unit **220** moves backward.

(82) In the case of FIG. 6A, when the expansion fixing unit **220** is completely inserted and fixed into the communication port **112**, the outer side surface of the expansion fixing unit **220** is in the state of being in close contact with the inner surface of the communication port **112**, and at this time, the communication port **112** is in a state of being blocked by the expansion fixing unit **220**.

(83) However, as shown in FIG. 6B, when a part of the expansion fixing unit **220** is unintentionally discharged to the outside of the communication port **112** with a preset length or more, the decompression part **140** is exposed to the outside, and the high pressure gas in the container **10** is discharged to the outside.

(84) At this time, in the case of the container **10**, the high pressure is maintained because the high pressure gas is supplied from the air channel. Accordingly, when the expansion fixing unit **220** slides from and is pushed out the communication port **112** over a certain length, the high pressure gas is discharged and the pressure inside the container **10** decreases.

(85) If the decompression unit **140** does not exist, the discharge module **200** may pop out unintentionally due to the high pressure inside the container **10**.

(86) Of course, when the fixing module **300** is coupled to the inclined protruding unit **110** and the locking protrusion **113** is fastened to one end portion of the adjusting unit **320**, the discharge module **200** is supported not to pop out. Even if the adjusting unit **320** is unintentionally damaged or the coupled state is released, the decompression part **140** can prevent the discharge module **200** from popping out.

(87) In other words, if the gas pressure inside the container **10** is too high, or if the expansion fixing unit **220** is partially pushed out of the communication port **112** due to damage of the fixing module **300**, by discharging the gas from the decompression part **140** to the outside so that the exhaust module **200** does not pop out, it is possible to prevent a safety accident from occurring in the user.

(88) Further, the decompression part **140** according to the present invention may be formed in a slit shape as shown in FIG. 7.

(89) As shown in the drawing, the decompression part **140** is recessed in a slit shape at the upper end portion of the inclined protruding unit **110**, and is formed to be partially exposed and opened when the expansion fixing unit **220** slides rearward.

(90) At this time, the decompression part **140** has a configuration designed to have substantially the similar purpose and effect to the shape of FIGS. 6A and 6B but is only different from the position thereof, and can adjust the extruded length of the expansion fixing unit **220** until the high pressure gas is discharged to the outside according to the slit-shaped length of the decompression part **140**.

(91) Also, the inclined protruding portion **110** can be formed to include both the hole shape shown in FIGS. 6A and 6B and the slit-shaped decompression portion **140** shown in FIG. 7.

(92) As described above, the inclined protruding unit **110** according to the present invention preemptively discharges the gas inside the container **10** when the expansion fixing unit **220** is pushed out from the communication port **112** by a preset length or more through the decompression part **140**. Accordingly, it is possible to prevent the discharge module **200** from unintentionally detaching due to the high pressure inside the container **10**, thereby suppressing a safety accident.

(93) While preferred embodiments of the present invention have been shown and described above, the present invention is not to be limited to the particular embodiments described above, but rather can be modified without departing from the scope of the present invention as claimed in the appended claims. It goes without saying that various modifications can be made by those who have ordinary knowledge in the technical field to which the present invention belongs, and such modifications should not be understood separately from the technical idea or perspective of the present invention.

DESCRIPTION OF SYMBOLS

(94) **10**: Container **12**: Housing space **100**: Guide module **110**: Inclined protruding unit **120**: Cover unit **130**: Protrusion pressing part **140**: Decompression part **200**: Discharge module **210**: Channel unit **220**: Expansion fixing unit **230**: Connecting unit **300**: Fixing module **310**: Coupling unit **320**: Adjusting unit

Claims

1. A cover device which is coupled to an opening portion of a container in which a photoresist aqueous solution is housed to seal the inside and simultaneously discharge the photoresist aqueous solution to the outside, the cover device comprising: a guide module which is coupled to an upper end portion of the container, has an upward inclination at a predetermined angle, has an inclined protruding unit that protrudes, and is formed with a communication port through which the inside and the outside of the housing space communicate; a discharge module which is formed in a long pipe shape, and in which one side along a longitudinal direction penetrates the communication port and is disposed inside the container, the other side is exposed to the outside to form a discharge port, and an intermediate portion along the longitudinal direction has an expansion fixing unit which is formed to expand, is in contact with an inner surface of the communication port on at least a part of periphery, and is inserted and fixed; and a fixing module which is fixedly coupled to the inclined protruding unit in a state in which the other part of the discharge module penetrates, and is formed so that the discharge module is pressed toward the guide module, wherein the communication port is formed to have an inner diameter that decreases toward the inside of the container in at least a partial region and is fixed by pressing.
2. The cover device according to claim 1, wherein the discharge module further comprises a channel unit which is formed long, has a part disposed inside the container, and has a discharge channel formed therein, and a connecting unit which protrudes to extend from the expansion fixing unit, is exposed to outside, and has a discharge port extending with the discharge channel.
3. The cover device according to claim 2, wherein the inclined protruding units further comprises a protrusion pressing part formed in at least a part along the inner surface of the communication port, protrudes toward the center, and comes into contact with the expansion fixing unit.
4. The cover device according to claim 3, wherein the protrusion pressing part is formed with a first inclined surface that protrudes at a predetermined angle along the longitudinal direction.
5. The cover device according to claim 4, wherein the inclined protruding unit is formed with a second inclined surface, which is continuous with the first inclined surface and is inclined in an opposite direction inside the communication hole, and has a relatively larger inclination angle than the first inclined surface.
6. The cover device according to claim 2, wherein the inclined protruding unit further comprises a decompression part which is formed in at least a part along the periphery, and when the expansion fixing unit is pushed out from the inside of the communication port by a preset length or more and slides toward the connecting unit, discharges the gas inside the container to the outside.
7. The cover device according to claim 6, wherein the decompression part is formed in a hole shape through which inside and outside communicate in at least a part along the periphery of the communication port, and is formed to be opened when the expansion fixing unit moves rearward.
8. The cover device according to claim 6, wherein the decompression part is formed in a slit shape at an upper end portion of the inclined protruding unit, and opened when the expansion fixing unit moves rearward.
9. The cover device according to claim 2, wherein the fixing module comprises: a coupling unit, through which the other part of the discharge module penetrates, and which surrounds at least a part of the inclined protruding unit in a state of being in contact with the expansion fixing unit; and an adjusting unit which has a fixed length, and an intermediate portion fixed to the coupling unit,

and in which one side along the longitudinal direction selectively comes into contact with the inclined protruding unit to adjust coupling with the coupling unit, wherein the adjusting unit is operated to press the expansion fixing unit in the direction of the container, and the expansion fixing unit is prevented from being detached in an opposite direction.

10. The cover device according to claim 9, wherein the inclined protrusion unit comprises a serrated locking protrusion on at least a part of the outer surface, and a plurality of protrusions are continuously formed along the protruding longitudinal direction, and the adjusting unit is coupled to at least a part of the locking protrusion on one end portion, and is selectively fixed.
